

SHARMILA BALASUBRAMANIAN

BRAMPTON ON, CANADA | 571-551-9500 | sharmila71990@gmail.com |

[LinkedIn: sharmila-balasubramanian](#)

SENIOR FRONTEND ENGINEER | REACT | TYPESCRIPT | AI-DRIVEN PRODUCT ENGINEERING

Senior Frontend Engineer with 10+ years of experience building scalable React and TypeScript applications, AI-assisted engineering workflows, reusable UI systems, and enterprise web platforms supporting 100M+ users across ecommerce, fintech, and enterprise environments. Strong expertise in frontend architecture, modern JavaScript engineering, scalable component systems, API-driven applications, developer productivity, performance optimization, and AI-assisted software development practices. Proven track record modernizing legacy frontend ecosystems into scalable React architectures, improving Core Web Vitals by 30%, accelerating engineering delivery through reusable platform infrastructure, and enabling faster product iteration using modern engineering workflows. Experienced collaborating cross-functionally with product, backend, DevOps, QA, and design teams to deliver high-performance customer experiences in Agile environments. Passionate about AI-powered engineering workflows, scalable frontend platforms, developer experience, and building modern user-centric digital products.

TECHNICAL SKILLS

Frontend Technologies : React.js, JavaScript (ES6+), TypeScript, Angular, HTML5, CSS3, Responsive Web Design, SPA Architecture

AI & Modern Engineering : AI-Assisted Development, Prompt-Driven Engineering Workflows, Intelligent UI Experiences, Productivity Automation, AI-Augmented Software Development

Frontend Architecture & Product Engineering : Reusable Component Libraries, Design Systems, Frontend Architecture, Scalable UI Platforms, Product Engineering, Developer Experience Improvements

React Ecosystem : React Hooks, Redux, Context API, State Management, Component-Driven Architecture

Backend & APIs : Node.js, Express.js, REST APIs, GraphQL, API Integration, Asynchronous Programming, JSON

Performance Optimization : Core Web Vitals, Lazy Loading, Code Splitting, Rendering Optimization, Memoization, Frontend Scalability

Testing & Quality Engineering : Jest, React Testing Library, Cypress, Playwright

DevOps & Developer Tooling : Git, CI/CD Pipelines, Jira, Splunk, Dynatrace, Feature Flags, Debugging, Production Support

Engineering Practices : Agile/Scrum, SDLC, Code Reviews, Technical Documentation, Application Modernization, AI-Assisted Development

WORK EXPERIENCE

Senior Frontend Developer – React | Product Engineering Lead

Walmart International, Toronto, ON

Nov 2023 – Present

- Led development of scalable React.js and TypeScript applications supporting 100M+ monthly users across global ecommerce platforms.
- Built reusable frontend platforms and scalable component systems improving engineering consistency and reducing frontend development effort by 25%.
- Improved Core Web Vitals and frontend application performance by 30% through rendering optimization, lazy loading, code splitting, and efficient state management strategies.
- Collaborated closely with backend engineering teams integrating enterprise APIs and Node.js-based services to deliver scalable customer-facing experiences.
- Contributed to AI-assisted engineering workflows and modern development practices improving developer productivity and delivery efficiency.
- Developed scalable frontend architecture patterns enabling faster feature delivery and reusable implementation across distributed engineering teams.
- Implemented automated testing strategies using Jest and React Testing Library, improving release quality and reducing regression issues.
- Led frontend code reviews and contributed to engineering standards focused on scalability, maintainability, accessibility, and performance optimization.
- Supported CI/CD-driven deployments, monitoring, debugging, and production issue resolution within fast-paced Agile delivery environments.
- Partnered cross-functionally with product managers, UX designers, QA teams, and platform engineers to deliver customer-focused digital experiences.

Senior Front-End Engineer – UI Architecture & Strategy

Walmart Canada (via Publicis Sapient), Toronto, ON

Jul 2019 – Jun 2022

- Architected and delivered enterprise-scale React applications emphasizing scalability, maintainability, reusable infrastructure, and modern frontend engineering practices.
- Modernized legacy frontend ecosystems into scalable React-based platforms, reducing technical debt and improving engineering velocity.
- Built reusable UI modules, shared component libraries, and scalable frontend foundations accelerating feature delivery across multiple engineering teams.
- Collaborated with backend and platform engineering teams integrating enterprise APIs and scalable ecommerce workflows.
- Improved developer productivity through reusable architecture patterns, CI/CD workflows, and frontend engineering standardization.
- Participated in architecture discussions, sprint planning, technical strategy sessions, and cross-functional Agile collaboration initiatives.
- Contributed to frontend engineering documentation, reusable code standards, and scalable UI platform practices.
- Worked within data-driven product environments focused on customer experience optimization and scalable digital commerce solutions.

Senior Software Developer

Publicis Sapient (Client: Intel), Toronto, ON

Feb 2019 – Jun 2019

- Migrated legacy AngularJS applications into scalable modern frontend frameworks improving maintainability, performance, and developer efficiency.
- Built reusable UI components and modular frontend systems accelerating engineering delivery cycles.
- Improved application responsiveness and rendering efficiency through frontend optimization initiatives.
- Collaborated with distributed engineering teams supporting enterprise modernization and frontend platform initiatives.

Front-End Developer

Toronto-Dominion Bank (via Wipro), Toronto, ON

Aug 2018 – Jan 2019

- Developed enterprise banking applications using Angular, TypeScript, JavaScript, HTML5, and CSS3.
- Worked closely with stakeholders and developers to gather requirements and deliver scalable frontend solutions.
- Participated in debugging, testing, and production support activities for mission-critical banking applications

Technology Analyst

Infosys (Client: Verizon), India

Dec 2016 – Dec 2017

- Redesigned UI using AngularJS, improving performance and usability
- Ensured cross-browser compatibility and accessibility compliance
- Participated in code reviews and maintained development standards

KEY ACHIEVEMENTS

- Delivered scalable enterprise web platforms supporting 100M+ users across ecommerce and fintech ecosystems.
- Improved Core Web Vitals and frontend performance metrics by up to 30% across enterprise applications.
- Reduced frontend engineering effort by 20–25% through reusable component systems and scalable UI architecture.
- Increased release quality and reduced production defects through automated testing and CI/CD-driven workflows.
- Led frontend modernization initiatives transforming legacy applications into scalable React ecosystems.
- Contributed to engineering excellence through mentoring, architecture discussions, reusable platform design, and frontend best practices.
- Supported AI-assisted development practices improving engineering productivity and accelerating feature delivery.

EDUCATION

Bachelor of Computer Science Engineering

Sastra University

Additional Information

- Canadian Citizen
- Experience working in Agile/Scrum environments with distributed teams
- Strong communication, collaboration, and stakeholder management skills
- Passionate about AI-driven product engineering, scalable frontend platforms, developer productivity, and high-performance web applications